

MAGNN: Multi-scale adaptive graph neural networks with contrastive learning for malicious network traffic detection

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ABSTRACT

Network traffic analysis is a critical task in cybersecurity, enabling the classification of malicious traffic and the identification of potentially dangerous connections. Existing approaches often utilize traditional graph neural networks (GNNs) to represent and analyze the inherent graph-like structures in network traffic, where nodes correspond to entities such as IP addresses or devices, and edges represent communication or data flow between them. However, these methods primarily focus on node features, often neglect crucial edge-level (packet-level) information and struggle to adapt to diverse and evolving traffic patterns. To overcome these limitations, we propose a novel framework, Multi-Scale Adaptive Graph Neural Networks (MAGNN), for malicious network traffic detection. MAGNN integrates both node and edge features within a multi-scale contrastive learning framework, introducing three novel mechanisms: temporal-node contrast, edge-level contrast, and multi-head hierarchical contrast, to capture complex dependencies in network behavior. MAGNN is evaluated on four diverse benchmark datasets-CTU-13, ISCXVPN2016, CICIDS-2017, and CIRA-CIC-DoHBrw-2020-demonstrating superior performance in detecting malicious traffic. In addition to classification accuracy, MAGNN supports temporal traffic trend estimation, where error-based (regression-task) metrics are employed to assess the model's ability to forecast evolving traffic dynamics. Furthermore, MAGNN demonstrates robust scalability through linear computational complexity, support for distributed training with multi-GPU architectures, and efficient convergence across large-scale graphs.

1. Introduction

The increasing frequency and sophistication of cyberattacks pose substantial threats to the security of computer networks and information systems. Malicious network traffic detection methods are critical for protecting critical infrastructure by analyzing raw network traffic, such as packet captures or flow-based records, to extract detailed statistics and identify potential threats. Traditional malicious detection approaches are broadly categorized into signature-based and behavior-based methods. Signature-based methods rely on predefined rules or metrics to classify traffic, while behavior-based approaches use machine learning to detect complex and evolving attack patterns [1]. Behavior-based approaches [2,3] are preferred for their ability to detect previously unseen attacks, including zero-day vulnerabilities, but their reliance on large volumes of labeled data often limits their practicality in real-world scenarios [4].

Another limitation of traditional detection systems is their inability to capture topological patterns in network traffic. These patterns are vital for identifying advanced intrusions, such as persistent threats that involve lateral movement across a network [5]. Incorporating topological relationships into detection frameworks can significantly enhance their ability to detect sophisticated attacks by analyzing network graph structures and communication pathways [6].

Graph-based deep learning has emerged as a promising approach for addressing these challenges due to its ability to handle complex relational data [7,8]. Within this domain, graph-based malicious traffic detection identifies malicious nodes in graph structures and has been widely applied in areas such as financial fraud detection, misinformation detection, and network intrusion detection [9,10]. Unlike other fields of malicious detection, graph-based methods uniquely combine node features and graph structures, enabling a more comprehensive analysis [11]. Malicious traffic is often categorized as either

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feature-based, where node attributes deviate significantly from their neighbors, or structural, where anomalous patterns emerge in closely connected groups of nodes [12].

Over the years, numerous methods have been developed to address these challenges. Early approaches primarily compared a node's features with its neighbors or used network structure for detection [13,14]. Other techniques focused on feature subspace selection [15,16], but these methods often relied on domain-specific knowledge and failed to capture deep, non-linear relationships in graph data, limiting their scalability and effectiveness.

The advent of Graph Neural Networks (GNNs) [17] marked a major breakthrough in graph-based malicious traffic detection. Early GNN-based approaches, such as Anomal-E [18], identified malicious traffic by comparing restructured features and adjacency matrices. However, these methods often failed to capture critical malicious patterns due to their reliance on neighborhood aggregation, which can dilute essential signals [19]. Recent advancements introduced contrastive learning paradigms [20] to graph-based malicious detection, leveraging relationships between nodes and their neighborhoods to improve representation learning [21]. While effective, these methods often neglect subgraph-level information, which is crucial for optimizing embeddings and capturing complex dependencies in graph data [22].

To address these limitations, we propose MAGNN: Multi-Scale Adaptive Graph Neural Networks for malicious traffic detection. MAGNN combines node and edge features with the graph's topological structure in a multi-scale contrastive learning framework. The framework introduces three innovative mechanisms-temporal-node contrast, edge-level contrast, and multi-head hierarchical contrast-to capture diverse dependencies and improve detection accuracy. This multi-view design integrates various sources of malicious traffic information, computing reliable malicious scores for each node while maintaining scalability and robustness. Furthermore, MAGNN incorporates graph augmentation techniques to enhance subgraph representation learning, addressing challenges commonly encountered in practical deployment, such as data scarcity and evolving attack patterns.

Unlike traditional supervised models, MAGNN operates in a self-supervised manner, reducing its reliance on extensive labeled data and making it more practical for deployment in operational network environments where labeled traffic is often unavailable. By leveraging the graph-based nature of network traffic-where nodes represent hosts and edges denote communication flows-MAGNN captures both node and edge features, enabling the detection of sophisticated intrusions. To address the practical challenges posed by contemporary network environments-particularly the scarcity of labeled malicious traffic-MAGNN is designed around a self-supervised training paradigm. Instead of relying on large labeled datasets, MAGNN learns informative representations by optimizing multi-scale contrastive objectives derived from intrinsic graph structures and temporal dynamics. These include temporal-node, edge-level, and hierarchical contrasts, which allow the model to extract semantic features from unlabeled traffic data. During evaluation or deployment, a small set of labeled examples can optionally be used for fine-tuning or scoring, making MAGNN highly adaptable to semi-supervised scenarios. This hybrid capability ensures that the framework remains robust, scalable, and effective in modern cybersecurity operations where labeling is often limited or delayed. The proposed MAGNN framework makes the following key contributions to the field of malicious network traffic detection:

1. MAGNN introduces an innovative multi-scale contrastive learning framework that integrates temporal-node, edge-level, and multi-head hierarchical contrasts. This approach enables the model to capture diverse malicious patterns across varying levels of granularity, significantly enhancing its ability to detect both localized malicious activities and broader structural deviations in network traffic.
2. The framework employs a graph augmentation technique that combines edge modification and random walk with restart to generate

complementary graph views. This strategy simulates realistic variations in network traffic, enhancing the model's robustness and adaptability to emerging attack scenarios and diverse malicious behaviors.

3. Unlike traditional graph neural network methods, which primarily focus on node-level features, MAGNN explicitly incorporates temporal information and edge-level attributes. This comprehensive integration addresses the limitations of node-centric approaches and provides a holistic understanding of network flow patterns, critical for detecting sophisticated malicious traffic.
4. MAGNN adopts a self-supervised learning strategy that enables training without labeled data, while supporting optional semi-supervised refinement during evaluation. This design ensures robust performance on large-scale datasets, maintains computational efficiency, and enhances applicability in practical cybersecurity environments where labeled data is limited.

The rest of the paper is organized as follows. Section 2 discusses related work in network traffic detection. Section 3 details the proposed model architecture and design. Section 4 presents the experimental setup and evaluation metrics. Section 5 discusses the results and comparisons with baseline methods. Finally, Section 6 concludes the paper with future research directions.

2. Related work

The detection of malicious network traffic has been a crucial area of research, with significant advancements in leveraging graph-based models and deep learning techniques to address increasingly sophisticated cyber threats. Early approaches, including those by Li et al. [23] and Zola et al. [24], employed traditional graph-based methods that primarily analyzed the structural and feature-based properties of network graphs. While these methods offered initial insights into malicious behavior, their limitations in capturing complex, non-linear dependencies and adapting to dynamic network behaviors hindered their effectiveness in evolving threat scenarios.

The emergence of GNNs has revolutionized the field by enabling the modeling of intricate dependencies within graph-structured data. GNN-based approaches have demonstrated significant improvements in identifying malicious activities by leveraging both local and global graph features. For instance, Zhou et al. [25] introduced a GNN-based framework for botnet detection that focused on the topological structure of network traffic, achieving higher accuracy than traditional machine learning methods. Similarly, Liu et al. [12] proposed a Contrastive self-supervised Learning framework for Anomaly (CoLA), a contrastive learning-based technique for anomaly detection in graph data, which inspired further refinements like the Abnormality-Aware Graph Neural Network (AAGNN) [25] and Host-based Intrusion Detection System (HIDS) [26]. These advancements incorporated sophisticated paradigms, including one-class GNNs and GCN-based feature extraction, to enhance the robustness of malicious traffic detection.

The integration of self-supervised learning into GNN frameworks has further expanded their capabilities, particularly in handling unlabeled data-a common challenge in real-world cybersecurity applications. Anomal-E [18], for example, utilized self-supervised methods to generate edge embeddings and detect network intrusions, outperforming traditional feature-based models. Similarly, the adoption of graph contrastive learning methods, as highlighted by Velickovic et al. [27], Li et al. [28], and Hafdi et al. [29], has been instrumental in improving malicious traffic detection. These methods employ techniques such as node-subgraph and edge-level contrasts to optimize node representations and identify malicious patterns more effectively.

Data augmentation techniques have also been integrated into graph-based frameworks to improve generalization and robustness. Strategies such as Robust Self-Aligned (RoSA) [30] and Random Walk with Restart (RWR) [31] create augmented graph views, simulating realistic variations in network traffic. GGraph Anomaly Detection AugmenTED

(GRADATE) [32] further advanced this field by combining graph augmentation with multi-scale contrastive learning, capturing relationships across nodes, subgraphs, and the entire graph. These methodologies enhance the detection of diverse and evolving attack patterns, showcasing the potential of graph augmentation in malicious traffic detection.

Despite these advancements, many existing models rely heavily on node-level features and overlook the importance of integrating edge-level information and temporal dynamics, which are critical for a comprehensive understanding of network flow patterns. The Net-flow Edge Graph Attention Network (NEGAT) [17], for instance, addressed this gap by incorporating Net Flow-based features to distinguish network attacks in real-time. However, the scalability and adaptability of such approaches remain limited in dynamic and large-scale network environments.

Recently, the Robust Adaptive Graph Network (RAGN) [33] introduced a dynamic attention mechanism and iterative refinement of graph structures and node features, guided by feature smoothness regularization. This design allows RAGN to adapt to adversarial changes in real time by assigning lower attention weights to unreliable edges and refining node features to suppress perturbations. These mechanisms collectively enhance its robustness and accuracy in dynamic, adversarial environments. In contrast, the proposed MAGNN framework adopts a multi-scale self-supervised contrastive learning strategy that captures both localized and global structural semantics within graph-structured network traffic data. Instead of relying on explicit regularization, MAGNN leverages hierarchical contrastive objectives-including temporal-node, edge-level, and subgraph-to-graph contrast-guided by multi-head attention mechanisms to enhance the expressiveness and robustness of node representations. Furthermore, it incorporates graph augmentation techniques to enrich subgraph learning and improve generalization under limited data and evolving attack patterns, making it well-suited for dynamic malicious traffic detection tasks across diverse real-world scenarios.

Recent efforts by Zhou et al. introduced a Reconstructed Graph neural network with Global-Local Distillation (RG-GLD) [34], which focuses on building lightweight GNNs for anomaly detection by reducing model complexity through knowledge distillation. Unlike this approach, MAGNN emphasizes multi-scale semantic representation learning via contrastive objectives, providing robustness across diverse attack types while maintaining computational efficiency through subgraph sampling and distributed training.

Li et al. [35] proposed Flow GAN Anomaly, an adversarial learning-based anomaly detection framework that models traffic flow distributions to identify anomalies in network intrusion detection systems. While effective in capturing distributional shifts, such generative approaches often face limitations in interpretability and scalability. In contrast, MAGNN explicitly models structural and semantic dependencies through multi-scale contrastive learning on graphs, enhancing both generalization and explainability.

The Flow Interaction Relationship Graph Neural Network (FIR-GNN) [36] leverages flow interaction relationships to detect intrusions in consumer IoT environments using graph neural networks. Our work differs by aiming for broader generalization across multiple domains and by integrating both temporal and edge-level contrastive mechanisms. This design enables MAGNN to identify malicious behaviors across a wider range of traffic environments, beyond smart homes and into more diverse scenarios. Building on these foundations, our proposed MAGNN framework introduces a multi-scale contrastive learning approach that incorporates temporal-node, edge-level, and multi-head hierarchical contrasts. By integrating node and edge features with the graph's topological structure, MAGNN addresses the limitations of existing methods and enables the detection of both localized and global malicious behaviors. Furthermore, the framework leverages advanced graph augmentation techniques, including edge modification and RWR, to enhance its robustness and adaptability to diverse attack scenarios.

Table 1 provides a comparative summary of key challenges in existing malicious traffic detection approaches and illustrates how the proposed MAGNN framework effectively addresses these limitations.

3. Design of MAGAN model

3.1. Preliminaries

We define the task of malicious traffic detection within the framework of graph-based analysis, where the network traffic is represented as an undirected graph $G = (V, E)$. Here, $V = \{v_1, v_2, \dots, v_N\}$ represents the set of N nodes, each corresponding to a network entity such as an IP address or a device, and E denotes the set of M edges, representing interactions or connections between these entities, such as network packets or communication sessions.

Each node $v_i \in V$ is associated with a d -dimensional feature vector encapsulated in the node feature matrix $X \in \mathbb{R}^{N \times d}$. These features may include attributes such as traffic volume, packet statistics, or behavioral patterns. The structural relationships among the nodes are captured by the adjacency matrix $A \in \mathbb{R}^{N \times N}$, where $A_{ij} = 1$ if an edge exists between nodes v_i and v_j , and $A_{ij} = 0$ otherwise. This adjacency matrix encodes the graph's topology, which reflects the inherent relationships within the network traffic data.

The objective of malicious traffic detection is to learn a scoring function $f : V \rightarrow \mathbb{R}$ that assigns a maliciousness score S_i to each node v_i . A higher score S_i indicates a greater likelihood that the node v_i is associated with malicious activity. The scoring function f must effectively leverage both the node-level attributes and the edge-level relationships within the graph. Moreover, it must capture the multi-scale dependencies within the network, integrating topological information and feature representations to identify malicious behaviors with high accuracy and robustness. This formulation lays the foundation for graph-based malicious traffic detection, emphasizing the critical role of both structural and feature-based representations in accurately modeling network traffic and identifying malicious entities (Table 2).

3.2. MAGNN framework

The proposed MAGNN framework is designed to address the complexities of malicious network traffic analysis through a multi-stage graph-based learning pipeline, as shown in Fig. 1. The process begins with constructing graph-structured representations from raw packet flows, where nodes encode diverse semantics such as temporal instances, protocol types, and edge-level time dependencies. The relationships among these nodes are captured in an adjacency matrix that preserves the topological structure of the network traffic, enabling MAGNN to effectively model interaction patterns.

MAGNN operates in a self-supervised learning setting using a contrastive learning objective. To support robust representation learning, MAGNN first generates two augmented views of the constructed traffic graph through stochastic transformations such as node dropout and topology-aware edge perturbations guided by graph centrality. These augmented views are passed through a shared graph encoder that computes node embeddings reflecting both structural and semantic similarities.

At the core of the MAGNN architecture is a multi-head hierarchical attention mechanism, which integrates information at three levels: (i) Node-level attention, focusing on the importance of immediate neighbors; (ii) Subgraph-level attention, capturing mid-range dependencies within semantic clusters; and (iii) Graph-level attention, aggregating global information across the entire graph to provide broader contextual understanding.

The representations learned at each level are passed through contrastive modules, which compute a contrastive loss between

Table 1
Comparative summary of addressed research gaps and contributions..

Challenge/Gap in prior work	Limitation in existing studies	How MAGNN addresses it
Static or non-adaptive graph construction.	Graphs are often predefined and fail to adapt to traffic dynamics.	MAGNN incorporates adaptive edge modification and sampling strategies to dynamically construct graphs from traffic data.
Lack of multiscale feature integration.	Prior methods often overlook the hierarchical nature of traffic features.	MAGNN fuses multi-scale features using hierarchical contrastive learning modules.
Vulnerability to adversarial perturbations.	Models can be easily misled by crafted or noisy samples.	MAGNN incorporates adversarial contrastive learning to improve robustness against perturbations.
Limited ability to generalize across datasets.	Models often overfit specific traffic types or sources.	MAGNN applies generalizable graph augmentation and contrastive objectives to improve cross-dataset performance.
High computational cost of full-graph GNNs.	Processing entire graphs increases latency and memory cost.	MAGNN uses subgraph sampling and multi-head attention to reduce computational overhead.

Table 2
Comprehensive list of abbreviations used throughout the paper to enhance clarity and understanding of technical terms..

Abbreviation	Description
GNN	Graph Neural Network
MAGNN	Multi-Scale Adaptive Graph Neural Network
CoLA	Contrastive self-supervised Learning framework for Anomaly
AAGNN	Abnormality-Aware Graph Neural Network
HIDS	Host-based Intrusion Detection System
GCN	Graph Convolutional Network
GRADATE	GRAph Anomaly Detection AugMENTed
NEGAT	Net-flow Edge Graph Attention Network
RAGN	Robust Adaptive Graph Network
RG-GLD	Reconstructed Graph neural network with Global-Local Distillation
FIR-GNN	Flow Interaction Relationship Graph Neural Network
RoSA	Robust Self-Aligned
EM	Edge Modification
RWR	Random Walk with Restart
MHHC	Multi-Head Hierarchical Contrast
MHA	Multi-Head Attention
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAPE	Mean Absolute Percentage Error
VPN	Virtual Private Network
CTU-13	Czech Technical University Botnet Dataset
ISCXVPN2016	Intrusion Detection Evaluation Dataset with VPN Traffic (2016)
CICIDS-2017	Canadian Institute for Cybersecurity Intrusion Detection Dataset (2017)
DoHBrw-2020	DNS over HTTPS Browsing Dataset (2020)
DDP	Distributed Data Parallel (PyTorch module)
APTs	Advanced Persistent Threats

corresponding nodes in the augmented views. This promotes invariance to perturbations while preserving meaningful discriminative features. The resulting embeddings are then evaluated using a malicious score function, which estimates the likelihood of a node or flow being associated with anomalous behavior.

This end-to-end pipeline-spanning graph construction, contrastive learning, hierarchical attention, and anomaly scoring-enables MAGNN to adapt dynamically to unseen attacks and evolving traffic patterns, while maintaining computational efficiency through distributed training and subgraph sampling strategies.

3.3. Graph augmentation

Graph augmentation plays a vital role in the self-supervised learning paradigm, enabling the model to extract deeper semantic information from the graph structure. In this paper, we employ edge modification and a random walk with restart to create a second view of the graph and sample subgraphs around nodes, respectively. These augmented graph views and subgraphs are fed into the graph contrastive network.

3.3.1. Edge modification

Edge Modification (EM) generates the second view of the graph by perturbing the edges in the adjacency matrix A . Inspired by [37], this

process involves both edge deletion and edge addition to maintain the graph's structural properties while ensuring robustness in the learned representations.

Let P represent the proportion of edges to modify, and M denote the total number of edges in the original graph. The number of edges to be deleted and added is defined as $\frac{PM}{2}$. The modified adjacency matrix A' is constructed as follows:

$$A' = A - A_{\text{delete}} + A_{\text{add}} \quad (1)$$

where A_{delete} represents the matrix with $\frac{PM}{2}$ edges uniformly and randomly removed, and A_{add} represents the matrix with $\frac{PM}{2}$ edges uniformly and randomly added.

This edge modification ensures that the second view retains critical graph properties while introducing sufficient variability for robust representation learning.

3.3.2. Random walk with restart

To effectively capture the local neighborhood structure around a target node, we use the RWR strategy [38] for subgraph sampling. This approach enables the selection of semantically and topologically relevant nodes based on their proximity to the target node v_i in the graph. For a given target node v_i , the RWR process defines an iterative probability update over time steps t , governed by the following recurrence relation:

$$\mathbf{p}_t = (1 - \alpha)\mathbf{A}\mathbf{p}_{t-1} + \alpha\mathbf{e}_i \quad (2)$$

Here, $\alpha \in [0, 1]$ is the restart probability, which determines the likelihood of jumping back to the starting node v_i at each step. $A \in \mathbb{R}^{N \times N}$ denotes the row-normalized adjacency matrix, computed as $A = D^{-1}\hat{A}$, where $\hat{A} = A + I$ includes self-loops and D is the diagonal degree matrix. This normalization ensures that each row of A represents a valid transition probability distribution across neighboring nodes.

The initialization of the random walk is explicitly defined by setting $\mathbf{p}_0 = \mathbf{e}_i$, where $\mathbf{e}_i \in \mathbb{R}^N$ is a one-hot vector with a value of 1 at the index corresponding to the target node v_i , and 0 elsewhere. This initialization assigns the entire probability mass to the starting node, ensuring that the walk originates from v_i . At each subsequent step t , the probability vector $\mathbf{p}_t \in \mathbb{R}^N$ is updated using the row-normalized adjacency matrix A .

The iterative updates continue until the probability vector $\mathbf{p}_t$ converges to a stationary distribution, denoted $\mathbf{p}_\infty$. This distribution encodes the steady-state probabilities of visiting other nodes from v_i , effectively capturing their proximity and relevance. Nodes with higher stationary probabilities are considered more structurally and semantically related to the target and are thus included in the sampled subgraph. This proximity-aware subgraph sampling mechanism facilitates the identification of meaningful local contexts for contrastive learning. Moreover, deviations in feature similarity between the target node and its neighbors in the sampled subgraph can signal potential anomalies in the network traffic.

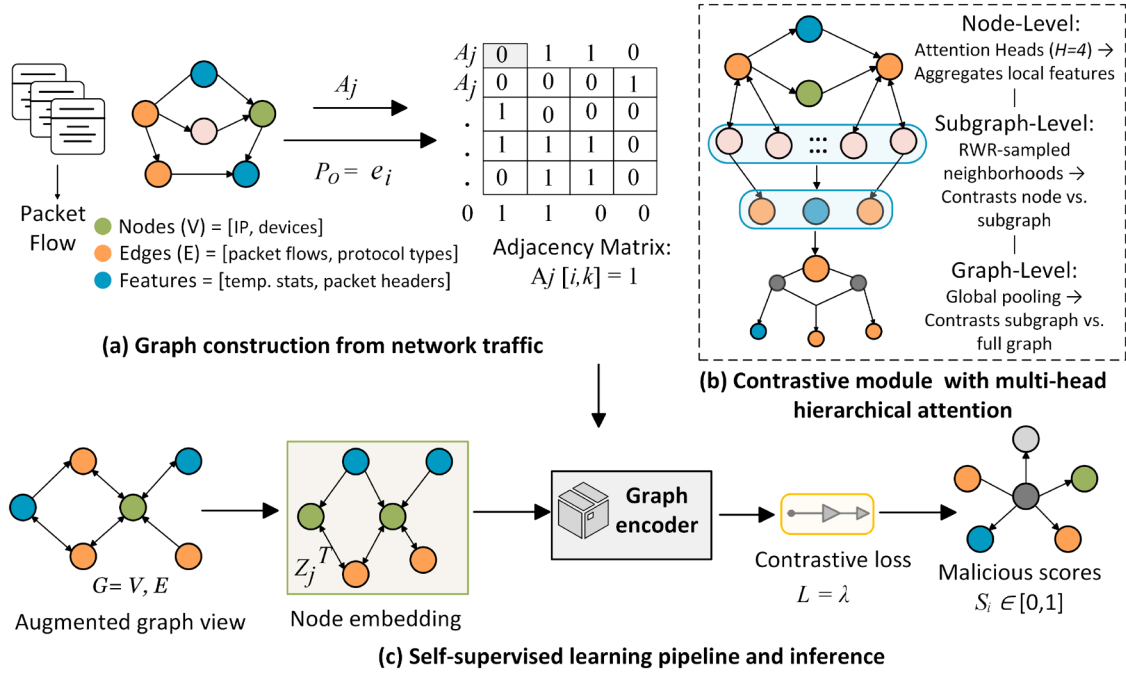


Fig. 1. Overview of the proposed MAGNN framework. (a) Network traffic is represented as graphs where nodes denote devices and edges represent packet flows. (b) Multi-head hierarchical contrastive module operates at the node, subgraph, and graph levels to learn robust feature representations. (c) Self-supervised learning process optimizes the model using contrastive loss, leading to accurate malicious score predictions for network entities.

3.4. Graph contrastive network

The contrastive learning paradigm has proven effective for graph anomaly detection [39]. In our proposed framework, the graph contrastive network operates on multiple views of the graph and incorporates three types of contrasts: temporal-node contrast, edge-level contrast, and multi-head hierarchical contrast.

3.4.1. Temporal-node contrast

Temporal-node contrast captures time-dependent anomalies by leveraging temporal features (e.g., packet timestamps) in network traffic. For a target node v_i , its temporal features are compared across different time intervals. The contrastive objective is defined as:

$$\mathcal{L}_{\text{temporal-node}} = - \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{z}_i^{(t_1)}, \mathbf{z}_i^{(t_2)})/\tau)}{\sum_{j \in \mathcal{N}} \exp(\text{sim}(\mathbf{z}_i^{(t_1)}, \mathbf{z}_j^{(t_2)})/\tau)} \quad (3)$$

where $\mathbf{z}_i^{(t_1)}$ and $\mathbf{z}_i^{(t_2)}$ are the embeddings of node v_i at time intervals t_1 and t_2 , $\text{sim}(\cdot, \cdot)$ denotes a similarity function (e.g., cosine similarity), τ is the temperature parameter, and $\mathcal{N}$ is the set of negative samples.

3.4.2. Edge-level contrast

Edge-level contrast focuses on detecting anomalies at the packet level by comparing edge embeddings across different views. For an edge e_{ij} between nodes v_i and v_j , the contrastive loss is defined as:

$$\mathcal{L}_{\text{edge-level}} = - \sum_{e_{ij} \in E} \log \frac{\exp(\text{sim}(\mathbf{z}_{e_{ij}}^{(1)}, \mathbf{z}_{e_{ij}}^{(2)})/\tau)}{\sum_{e_{kl} \in \mathcal{N}_E} \exp(\text{sim}(\mathbf{z}_{e_{ij}}^{(1)}, \mathbf{z}_{e_{kl}}^{(2)})/\tau)} \quad (4)$$

where $\mathbf{z}_{e_{ij}}^{(1)}$ and $\mathbf{z}_{e_{ij}}^{(2)}$ are edge embeddings in the two graph views, and $\mathcal{N}_E$ is the set of negative edge samples.

3.4.3. Multi-head hierarchical contrast

The Multi-Head Hierarchical Contrast (MHHC) module is designed to capture multi-scale structural dependencies by applying a Multi-Head Attention (MHA) mechanism across hierarchical levels of the graph—namely, nodes, subgraphs, and the entire graph [40]. This design allows the model to simultaneously focus on diverse relational patterns,

from localized node interactions to global graph-level structures, which is crucial for detecting both subtle and large-scale malicious activities in network traffic.

3.4.4. Multi-head attention mechanism

Before delving into the MHA mechanism, it is important to clarify the structure of the MHA mechanism, which is central to this module. The MHA allows the model to attend to different parts of the input graph simultaneously, capturing various types of interactions at different levels of granularity. Each attention head learns an independent similarity space, enabling the model to focus on different types of relations between nodes and subgraphs.

Key components of MHA

- **Number of attention heads (H):** This refers to the number of separate attention mechanisms used in parallel. Each attention head operates independently, attending to different features of the graph. The number of heads H is a hyperparameter that determines the model's ability to capture diverse relationships. In this design, we set H to a value that balances performance and computational efficiency.
- **Query, Key, and Value matrices (Q, K, V):** These matrices are used to project the input feature space into three different spaces: **queries (Q)**, **keys (K)**, and **values (V)**. For each attention head, the matrices $Q^{(h)}$, $K^{(h)}$, $V^{(h)}$ are linearly projected from the subgraph or node embeddings using learnable parameter matrices $W_Q^{(h)}, W_K^{(h)}, W_V^{(h)} \in \mathbb{R}^{d \times d'}$, where d is the input feature dimension and $d' = \frac{d}{H}$ is the dimensionality per head.
- **Scaled dot-product attention:** The attention scores are computed by measuring the similarity between the query and key vectors using the scaled dot-product attention mechanism. This allows each attention head to weigh the importance of different elements in the graph for a given query:

$$\text{Attention}(Q^{(h)}, K^{(h)}, V^{(h)}) = \text{softmax}\left(\frac{Q^{(h)}(K^{(h)})^T}{\sqrt{d'}}\right)V^{(h)} \quad (5)$$

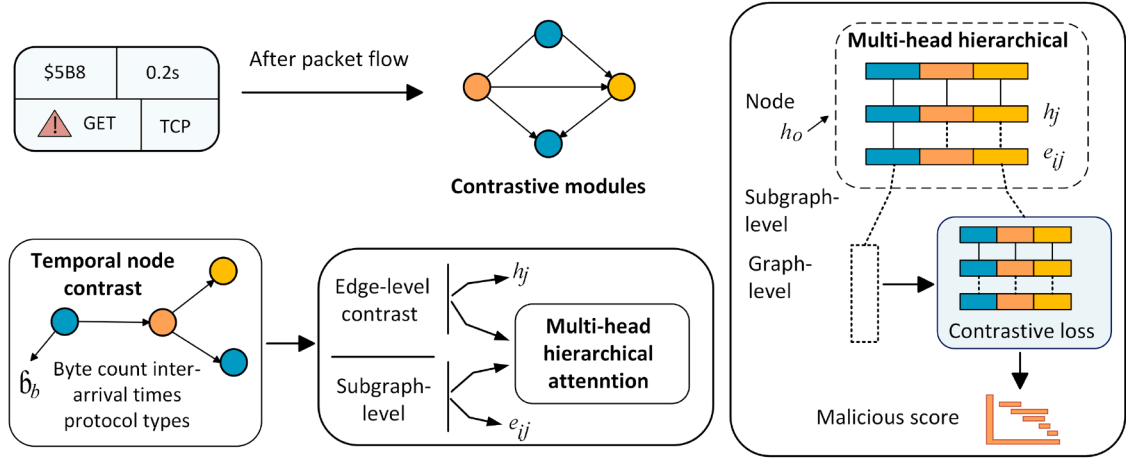


Fig. 2. MAGNN integrates temporal node, edge-level, and multi-head hierarchical attention modules, captures dependencies at nodes, subgraph, and global levels using multiple attention heads.

where $\sqrt{d'}$ is a scaling factor to stabilize the gradients during training.

- **Concatenation of outputs:** After the attention mechanism is applied in each head, the results from all heads are concatenated to form a unified representation:

$$\mathbf{z}_{S_i} = \text{Concat}(\mathbf{z}_{S_i}^{(1)}, \dots, \mathbf{z}_{S_i}^{(H)})W_O \quad (6)$$

where $W_O \in \mathbb{R}^{d \times d}$ is a learnable output projection matrix that combines the concatenated outputs into the final subgraph representation.

3.4.5. Multi-head hierarchical contrast loss

The MHHC loss utilizes the multi-head attention mechanism described above to impose a contrastive learning objective across multiple hierarchical levels of the graph. The attention-based hierarchical contrastive loss is defined as:

$$\mathcal{L}_{\text{MHHC}} = - \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathbf{z}_{S_i}, \mathbf{z}_G)/\tau)}{\sum_{j \in \mathcal{N}_S} \exp(\text{sim}(\mathbf{z}_{S_i}, \mathbf{z}_{S_j})/\tau)} \quad (7)$$

where: $\mathbf{z}_G$ is the embedding of the entire graph, $\mathcal{N}_S$ is the set of negative subgraph samples, $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity between embeddings, τ is a temperature scaling parameter that controls the sharpness of the probability distribution.

This loss encourages the model to bring similar subgraphs (positive samples) closer together in the embedding space while pushing dissimilar subgraphs (negative samples) apart, leading to improved generalization across hierarchical levels. The multi-head design in the MHA mechanism improves the model's capacity to generalize across varying structural contexts, enabling robust detection of both local anomalies and global threats in dynamic traffic graphs.

3.4.6. Integrated contrastive loss

The MAGNN framework combines three contrastive objectives to guide representation learning at different levels of graph structure:

$$\mathcal{L}_{\text{contrastive}} = \alpha_1 \mathcal{L}_{\text{temporal-node}} + \alpha_2 \mathcal{L}_{\text{edge-level}} + \alpha_3 \mathcal{L}_{\text{MHHC}} \quad (8)$$

where α_1 , α_2 , and α_3 are hyperparameters that balance the contribution of each contrastive component. This integrated contrastive loss constitutes the classification-driven objective in the MAGNN framework.

3.4.7. Multi-task loss function

MAGNN is formulated as a multi-task learning framework that simultaneously performs malicious traffic detection (classification) and temporal traffic prediction (regression). This joint optimization helps

the model to capture both static and dynamic characteristics of network traffic. The total training loss is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_1 \cdot \mathcal{L}_{\text{contrastive}} + \lambda_2 \cdot \mathcal{L}_{\text{regression}} \quad (9)$$

Here, $\mathcal{L}_{\text{contrastive}}$ is the integrated loss from Eq. 8, and $\mathcal{L}_{\text{regression}}$ is computed using MAE, MSE, RMSE, and MAPE between predicted and actual traffic feature values. λ_1 and λ_2 are task-level weighting coefficients used to balance the classification and regression objectives.

This multi-task loss enables MAGNN to learn temporally aware and semantically rich representations, improving its robustness in detecting evolving or stealthy attack patterns.

Attributed graph $G = (V, E, X)$, where: V is the set of nodes, E is the set of edges, $X \in \mathbb{R}^{|V| \times d}$ is the node feature matrix, where each row $\mathbf{x}_v \in \mathbb{R}^d$ represents a d -dimensional feature vector for node $v \in V$.

3.5. Optimization and complexity analysis

The proposed MAGNN framework is end-to-end trainable using back-propagation and gradient descent. During training, the model simultaneously minimizes the classification-driven contrastive loss and the auxiliary regression loss using the multi-task formulation in Eq. 9.

In terms of computational complexity, the time complexity of MAGNN is dominated by graph propagation and embedding transformation steps. As shown in Section 3.5.1, the model scales linearly with the number of nodes and edges, making it suitable for large-scale network traffic datasets.

By leveraging multi-scale contrastive learning objectives, our model can be efficiently optimized using gradient descent. The optimization process integrates the three contrastive losses- temporal-node contrast, edge-level contrast, and hierarchical contrast -into a single unified loss function as shown in Eq. 8.

3.5.1. Time complexity

The time complexity of the MAGNN framework is primarily influenced by two key operations: (i) neighborhood aggregation through graph neural network layers, and (ii) feature transformation via contrastive projection heads [41]. For a graph with $N = |V|$ nodes and $M = |E|$ edges, and assuming f_i is the dimensionality of the i -th layer and L is the total number of layers.

The first term, $M \sum_{i=0}^L f_i$, corresponds to the cost of message passing and attention computation across edges for each GNN layer. At each layer, messages (or neighbor features) are aggregated per edge, scaled by the dimensionality of the current layer's features.

$$\mathcal{O}\left(M \sum_{i=0}^L f_i + N \sum_{i=1}^L f_{i-1} f_i\right) \quad (10)$$

The second term, $N \sum_{i=1}^L f_{i-1} f_i$, represents the cost of feature transformation at each node, where node features are projected from f_{i-1} to f_i dimensions using learned linear transformations. Together, these two terms capture the computational load of the entire model during training. Importantly, this formulation confirms that the time complexity is linear with respect to the number of nodes and edges ($\mathcal{O}(N + M)$ when f_i is bounded), thus ensuring the MAGNN framework is scalable and efficient for processing large-scale network traffic graphs.

Algorithm 1 Proposed model: MAGNN framework.

Require: Attributed graph $G = (V, E, X)$, where V is the set of nodes, E is the set of edges, and $X \in \mathbb{R}^{|V| \times d}$ is the node feature matrix, where each row $\mathbf{x}_v \in \mathbb{R}^d$ represents a d -dimensional feature vector for node $v \in V$; number of training epochs T , batch size B , augmentation parameter P , restart probability α .

Ensure: Malicious score function $f(v)$ for each node $v \in V$.

- 1: **Initialization:** Randomly initialize the parameters of the MAGNN framework.
 - 2: **for** $t = 1$ **to** T **do**
 - 3: Generate two graph views by randomly deleting and adding $P \cdot |E|$ edges to simulate structural perturbation.
 - 4: Sample subgraphs for each node $v \in V$ using random walk with restart with restart probability α .
 - 5: Compute node embeddings $\mathbf{z}_v$ using a multi-layer GNN with multi-head attention.
 - 6: Compute edge embeddings $\mathbf{z}_e$ via concatenation and multi-layer perceptron on node pairs.
 - 7: Compute subgraph embeddings $\mathbf{z}_s$ using hierarchical attention pooling.
 - 8: Compute the temporal-node contrast loss $\mathcal{L}_{\text{temporal-node}}$ by comparing node embeddings across different time intervals.
 - 9: Compute the edge-level contrast loss $\mathcal{L}_{\text{edge-level}}$ by comparing edge embeddings across augmented graph views.
 - 10: Compute the multi-head hierarchical contrast loss $\mathcal{L}_{\text{MHHC}}$ using Eq. 7, capturing relationships between subgraph and global graph embeddings.
 - 11: Combine all contrastive losses into the integrated loss function using Eq. 8.
 - 12: Perform backpropagation and update the model parameters using gradient descent.
 - 13: **end for**
 - 14: Compute the malicious score $f(v) \in [0, 1]$ for each node $v \in V$ based on the learned embeddings and aggregated contrastive objectives.
 - 15: **Return** Malicious score function $f(v)$ for all nodes $v \in V$.
-

4. Experimental setup

4.1. Datasets description

We used four benchmark datasets to evaluate the effectiveness of the proposed MAGNN framework for malicious network traffic detection. These datasets represent diverse network traffic scenarios, attack types, and real-world network conditions.

1. The CTU-13 [42] dataset is a well-known benchmark dataset for botnet traffic analysis, developed by the Czech Technical University (CTU). It contains network traffic traces of both normal and malicious botnet activities across 13 distinct scenarios. The dataset includes a mix of real botnet infections, background traffic, and benign user activities, making it a valuable resource for evaluating the performance of intrusion detection and network anomaly detection systems. Its detailed annotations and realistic traffic patterns have made it a popular choice for research in cybersecurity and botnet detection.

2. The ISCXVPN2016 [43] dataset, also developed by the Canadian Institute for Cybersecurity, focuses on Virtual Private Network (VPN) traffic analysis. This dataset simulates realistic network traffic patterns, including both encrypted (VPN) and unencrypted traffic. It includes a variety of application types such as web browsing, Email, file transfer, and multimedia streaming, with malicious activities interspersed. The dataset is particularly valuable for evaluating the effectiveness of intrusion detection systems in handling encrypted traffic, which poses unique challenges for cybersecurity research.
3. The CICIDS-2017 [44] dataset, developed by the Canadian Institute for Cybersecurity, provides a comprehensive benchmark for intrusion detection systems. It includes network traffic data from both benign activities and a wide range of modern attack scenarios. This dataset simulates realistic day-to-day network traffic alongside malicious behavior, offering detailed features and diverse attack patterns. Its realism and scope make it an invaluable resource for evaluating the performance of malicious traffic detection models.
4. The CIRA-CIC-DoHBrw-2020 [45] dataset, created by the Canadian Institute for Cybersecurity, focuses on the detection of malicious traffic using DNS over HTTPS (DoH) communication. The dataset includes a variety of benign and malicious web traffic scenarios, representing a wide range of real-world browsing activities. Its detailed feature set captures the unique characteristics of encrypted DoH traffic, enabling researchers to evaluate the effectiveness of detection methods in handling this emerging and challenging network communication protocol.

4.2. Evaluation metrics

To comprehensively evaluate the performance of MAGNN, we employ both classification and error-based metrics.

Classification metrics: We use accuracy, precision, recall, and f1-score to assess the model's ability to correctly identify malicious and benign traffic. These metrics are computed using predicted class labels and are standard for evaluating binary and multiclass classification tasks. Precision and Recall are particularly important in cybersecurity contexts where minimizing false positives and false negatives is critical.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (11)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (12)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (13)$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

Error-based metrics: In addition to classification metrics, we report error-based metrics-Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE)-to assess the model's prediction confidence and misclassification cost. These metrics are calculated on the model's predicted probability scores (i.e., soft outputs from the final layer before thresholding). Their inclusion provides insight into how well the model's probabilistic outputs align with ground truth labels and helps quantify uncertainty or deviation from expected classification behavior.

Let $y_i \in \{0, 1\}$ denote the true label and $\hat{y}_i \in [0, 1]$ denote the predicted probability for the i -th sample, with n total samples. The formulas for the error metrics are as follows:

- **MAE** measures the average absolute difference between predicted probabilities and actual binary labels.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (15)$$

- **MSE** penalizes larger errors more heavily, reflecting overall prediction deviation.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (16)$$

- **RMSE** provides a more interpretable error scale by taking the square root of MSE.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (17)$$

- **MAPE** expresses the error as a percentage, offering an intuitive sense of prediction quality relative to actual values.

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i + \epsilon} \right| \quad (18)$$

where ϵ is a small constant added to avoid division by zero.

These error-based metrics complement traditional classification metrics by offering a finer-grained view of model calibration and misclassification impact, especially important in real-time applications where false confidence can be costly.

4.3. Implementation details

4.3.1. Experimental platform and software

All experiments were conducted on a Linux-based high-performance workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), 64 GB of RAM, and an AMD Ryzen 9 5950X CPU, running Ubuntu 22.04. The implementation was carried out using Python 3.9. We employed PyTorch 2.0 and PyTorch Geometric 2.3.1 as the primary frameworks for deep learning and graph-based processing. Data preprocessing and manipulation were performed using NumPy (v1.24) and Pandas (v1.5). This setup was chosen to ensure reproducibility, scalability, and compatibility with GPU-accelerated computation.

4.3.2. Model architecture and training setup

The MAGNN framework was developed using these open-source libraries to facilitate efficient deep learning computations and scalable graph processing. PyTorch Geometric was selected due to its robust support for graph operations and seamless integration with PyTorch. This enabled efficient implementation of the model's components, including the graph attention mechanisms, multi-scale contrastive learning modules, and hierarchical aggregation strategies. MAGNN learns network traffic representations through a combination of temporal-node, edge-level, and hierarchical contrastive learning objectives. Temporal dependencies were modeled using a graph attention mechanism with four attention heads and a hidden dimension size of 128, capturing sequential patterns while maintaining computational efficiency. The hierarchical structure was built using three graph convolutional layers, each processing up to two-hop neighborhoods to capture both local and global dependencies.

4.3.3. Training configuration and hyperparameters

To optimize model performance, hyperparameter tuning was conducted. The learning rate was set to 0.001 with a scheduler that decayed it by a factor of 0.1 after ten epochs without validation loss improvement. The Adam optimizer was used with a weight decay of 0.00001 to mitigate overfitting. A batch size of 64 was selected to balance memory and training efficiency. Training was performed over 100 epochs, with early stopping based on validation performance. Dirichlet neighbor sampling (size = 30) was adopted to ensure robustness in sparse environments and to reduce sensitivity to noise. Graph augmentation was applied through edge modification by randomly adding or deleting edges to generate diverse graph views for contrastive learning. This technique complemented the multi-scale contrastive learning

objective by encouraging the model to learn generalized and robust features. MAGNN was trained end-to-end using backpropagation with an integrated loss function that combined temporal-node, edge-level, and hierarchical contrastive components. The weighting coefficients λ_1 , λ_2 , and λ_3 were tuned empirically to balance the contributions of each contrastive mechanism. The overall setup ensured efficient scalability and robust performance across multiple datasets and traffic scenarios.

4.4. Baseline methods

- **Anomal-E** [18] employs a self-supervised approach to leverage edge features and graph topology for network intrusion detection. It generates edge embeddings without relying on labeled data, which can then be used with traditional malicious detection algorithms, such as isolation forests.
- **NEGAT** [17]: is used of NetFlow-based network traffic flows. The model integrates network flow insights into node representations, enhancing graph representation with self-supervision. It is designed to distinguish network attacks from hackers in real-time by monitoring network traffic flows across routers and identifying specific types of flows.
- **E-GraphSAGE** [46] leverages GNNs to detect and mitigate network traffic-based cyberattacks. By incorporating both edge features and topological information, it effectively models network traffic in a graph representation to identify malicious activities.
- **TCGNN** [47] is designed for packet-grained network traffic classification using GNNs. It transforms each network packet into an undirected graph and employs a two-layer graph convolutional network with three distinct aggregation strategies to enhance classification accuracy and performance.
- **DE-GNN** [48]: A graph-based structure that learns packet-level representations directly from raw bytes. It constructs network flows as a Traffic Interaction Graph, integrating behavioral patterns and packet-level features. It utilizes GNNs to extract flow-level features, providing a comprehensive view of network interactions.

5. Experimental results

5.1. Comparative performance evaluation against baseline methods on multiple datasets

The performance of the proposed MAGNN model is evaluated against several baseline methods across four benchmark datasets: CTU-13, IS-CXVPN2016, CICIDS-2017, and CIRA-CIC-DoHBrw-2020. Fig. 3 illustrates the accuracy comparison of MAGNN with Anomal-E, NEGAT, E-GraphSAGE, TCGNN, and DE-GNN under varying convergence steps. The results demonstrate that MAGNN consistently outperforms all baseline models across the datasets. For instance, on the CTU-13 dataset, MAGNN achieves an accuracy of approximately 97.80% at 100 convergence steps, surpassing all competitors. On the IS-CXVPN2016 dataset, it attains 98.60% accuracy, showing steady improvement throughout convergence and strong generalization capabilities. On the CICIDS-2017 dataset, MAGNN achieves an accuracy close to 99.13%, outperforming baseline models. Similarly, on the CIRA-CIC-DoHBrw-2020 dataset, the model reaches a peak accuracy of 99.30%, confirming its robustness and adaptability to diverse traffic patterns.

Fig. 4 shows the F1-score performance of MAGNN compared to the baselines under varying convergence steps. On the CTU-13 dataset, MAGNN achieves the highest F1-score of 98.94% at 100 steps, significantly outperforming models such as TCGNN and DE-GNN. On IS-CXVPN2016, it steadily improves to reach an F1-score of 99.36%, effectively addressing encrypted traffic challenges. For CICIDS-2017, MAGNN records an F1-score of approximately 98.97%, demonstrating its capacity to detect complex and diverse attack patterns. On CIRA-CIC-DoHBrw-2020, it achieves a peak F1-score of 99.10%, highlighting its adaptability to evolving network anomalies and malicious behaviors.

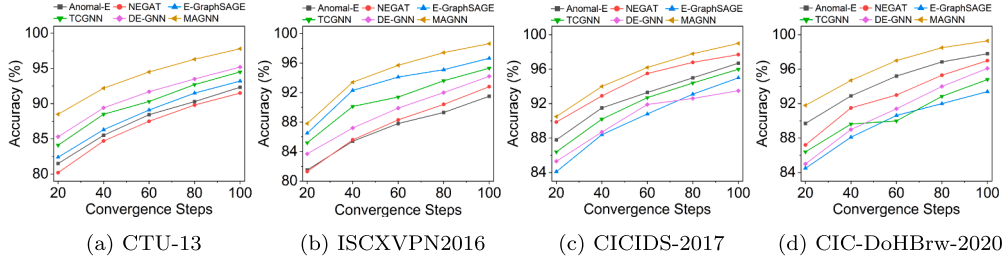


Fig. 3. Accuracy of different methods under varying convergence steps.

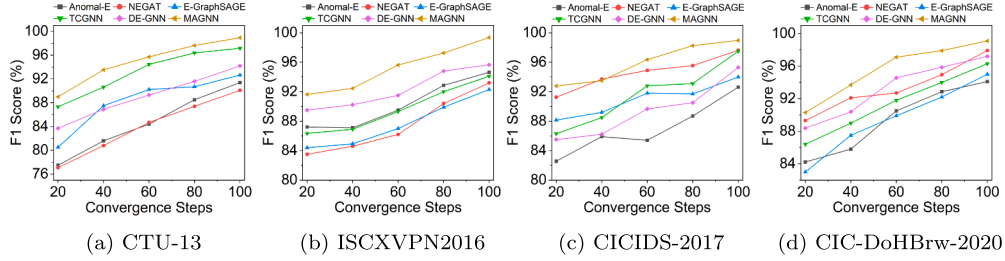


Fig. 4. F1 score of different methods under varying convergence steps.

These results validate the superiority of the MAGNN framework over existing methods. MAGNN consistently achieves higher accuracy and F1-scores across all datasets and convergence steps, reflecting faster convergence, improved classification performance, and strong generalization to varying traffic conditions. The consistent improvements further confirm its robustness and scalability. By integrating multi-scale adaptive graph learning and contrastive mechanisms, MAGNN offers a powerful and efficient solution for malicious network traffic detection.

Table 3 summarizes the performance of MAGNN compared to five baseline models across four benchmark datasets. On the CTU-13 dataset, MAGNN achieves the highest accuracy (98.45%), precision (97.88%), recall (97.30%), and F1-score (97.58%), along with the lowest error metrics (MSE: 0.06, RMSE: 0.20, MAE: 0.31, MAPE: 1.08). DE-GNN ranks second with an F1-score of 96.06%. For ISCXPVN2016, MAGNN maintains superior performance (Acc: 98.10%, F1: 97.12%) and low errors (MSE: 0.07, RMSE: 0.22). While TCGNN performs well, it falls short in classification accuracy and error reduction. On CICIDS-2017, MAGNN reaches an F1-score of 97.45% and again records the lowest error values. DE-GNN and TCGNN follow, but with weaker generalization. On the DoHBrw-2020 dataset, MAGNN attains the best results across all metrics (Acc: 99.20%, F1: 98.35%, MSE: 0.05), showcasing strong adaptability to complex traffic. E-GraphSAGE and DE-GNN are competitive but still lag behind. Overall, MAGNN consistently outperforms all baselines in both classification accuracy and error minimization. Its multi-scale adaptive graph learning and contrastive mechanisms enable robust generalization across diverse and evolving network traffic scenarios.

5.1.1. Analysis of model design relative to prior work

To contextualize the performance of the proposed MAGNN framework, we conducted a comprehensive comparison with several state-of-the-art models. As presented in Tables 3 to 5, MAGNN consistently outperforms all baseline methods across multiple datasets and a range of evaluation metrics, including error-based performance measures. Compared to models like Anomal-E and NEGAT, MAGNN's superior performance can be attributed to its multi-scale hierarchical design, which captures both local and global topological patterns. While E-GraphSAGE integrates edge features, it lacks hierarchical contrastive objectives, limiting its robustness to structural perturbations. DE-GNN and TCGNN attempt to capture deeper graph semantics, but they do not integrate explicit augmentation strategies, making them more sensitive to noisy or incomplete data.

In contrast, MAGNN combines edge modification, subgraph sampling, and contrastive learning at multiple levels (node, edge, and hierarchy), enabling it to learn more robust and generalizable representations of network traffic. These design elements collectively contribute to MAGNN's superior detection capabilities, especially in scenarios involving sparse or imbalanced traffic.

5.2. Network traffic prediction and analysis of prediction accuracy

Fig. 5 illustrates the MAE performance of all models across varying temporal steps on the four benchmark datasets. As expected, MAE values increase with longer prediction horizons, reflecting the growing difficulty of modeling long-term patterns. Nevertheless, the proposed MAGNN consistently achieves the lowest MAE across all datasets. On the CTU-13 dataset, MAGNN exhibits minimal error growth, consistently outperforming Anomal-E and NEGAT. Similarly, on ISCXPVN2016, MAGNN demonstrates significantly lower MAE values than TCGNN and DE-GNN. For CICIDS-2017, MAGNN maintains clear superiority over all baselines, particularly NEGAT and E-GraphSAGE. On CIRA-CIC-DoHBrw-2020, MAGNN sustains robust performance, achieving the lowest MAE throughout, indicating strong generalization to complex traffic patterns.

Fig. 6 presents MAPE values across increasing prediction steps. As a relative error metric, MAPE highlights prediction reliability. MAGNN consistently yields lower MAPE values across all datasets. On CTU-13, it maintains minimal error growth compared to E-GraphSAGE and Anomal-E. For ISCXPVN2016, MAGNN achieves the lowest MAPE at longer horizons, outperforming NEGAT and TCGNN. In the CICIDS-2017 scenario, MAGNN generalizes effectively, significantly reducing MAPE relative to DE-GNN and other baselines. Similarly, for CIRA-CIC-DoHBrw-2020, MAGNN sustains low and stable MAPE growth, reinforcing its resilience to evolving traffic patterns.

Fig. 7 displays MSE trends as temporal steps increase. As a squared error metric, MSE is particularly sensitive to larger deviations. MAGNN achieves the lowest MSE across all datasets. On CTU-13, it surpasses NEGAT, TCGNN, and E-GraphSAGE. On ISCXPVN2016, it consistently outperforms DE-GNN and TCGNN by a clear margin. For CICIDS-2017, MAGNN again records the lowest MSE, validating its accuracy and stability. On CIRA-CIC-DoHBrw-2020, it demonstrates outstanding error minimization, maintaining the best performance at all prediction horizons.

Table 3
Comparative performance of MAGNN and baseline methods on benchmark datasets..

Dataset	Model	Acc.	Pre.	Rec.	F1	MSE	RMSE	MAE	MAPE
CTU-13	Anomal-E	91.23	90.45	89.80	90.12	0.15	0.35	0.79	1.45
	NEGAT	93.75	92.90	91.87	92.38	0.12	0.31	0.72	1.38
	E-GraphSAGE	95.10	94.62	93.85	94.23	0.10	0.28	0.58	1.28
	TCGNN	96.45	95.82	94.93	95.37	0.09	0.26	0.50	1.22
	DE-GNN	97.12	96.34	95.78	96.06	0.08	0.24	0.42	1.15
	MAGNN	98.45	97.88	97.30	97.58	0.06	0.20	0.31	1.08
ISCXVPN2016	Anomal-E	90.90	89.87	88.65	89.25	0.16	0.38	0.81	1.50
	NEGAT	92.80	91.45	90.85	91.15	0.13	0.34	0.75	1.40
	E-GraphSAGE	94.50	93.60	92.90	93.25	0.11	0.30	0.61	1.30
	TCGNN	95.80	94.75	94.05	94.40	0.10	0.27	0.54	1.22
	DE-GNN	96.50	95.80	95.10	95.45	0.09	0.25	0.45	1.17
	MAGNN	98.10	97.45	96.80	97.12	0.07	0.22	0.33	1.10
CICIDS-2017	Anomal-E	92.35	91.50	90.90	91.20	0.14	0.32	0.77	1.42
	NEGAT	94.25	93.50	92.80	93.15	0.11	0.29	0.68	1.32
	E-GraphSAGE	95.65	94.80	94.15	94.45	0.10	0.26	0.55	1.25
	TCGNN	96.90	96.00	95.30	95.65	0.09	0.24	0.48	1.20
	DE-GNN	97.55	96.85	96.10	96.45	0.08	0.23	0.39	1.13
	MAGNN	98.65	98.00	97.40	97.70	0.06	0.19	0.29	1.05
DoHBrw-2020	Anomal-E	93.10	92.20	91.60	91.90	0.13	0.31	0.74	1.38
	NEGAT	95.00	94.10	93.50	93.80	0.10	0.28	0.66	1.28
	E-GraphSAGE	96.20	95.50	94.80	95.15	0.09	0.25	0.51	1.20
	TCGNN	97.35	96.55	95.95	96.25	0.08	0.24	0.43	1.15
	DE-GNN	98.00	97.25	96.70	97.00	0.07	0.22	0.38	1.10
	MAGNN	99.20	98.60	98.10	98.35	0.05	0.18	0.27	1.02

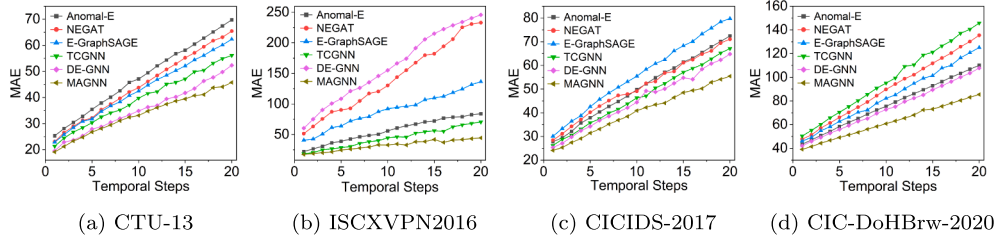


Fig. 5. MAE of different methods under varying prediction temporal steps.

Fig. 8 depicts RMSE values for MAGNN and the baseline models. As the square root of MSE, RMSE reflects the magnitude of overall prediction error. MAGNN again achieves the lowest RMSE across all datasets. On CTU-13, it demonstrates reduced RMSE growth, outperforming Anomal-E, NEGAT, and E-GraphSAGE. For ISCXVPN2016 and CICIDS-2017, MAGNN maintains clear advantages over all baselines, particularly under longer prediction horizons. On CIRA-CIC-DoHBrw-2020, it consistently achieves the lowest RMSE, highlighting its robustness and precision.

These results demonstrate that MAGNN significantly outperforms all baseline methods across MAE, MAPE, MSE, and RMSE metrics. Its superior performance is attributed to its multi-scale adaptive graph learning and contrastive mechanisms, which enable effective modeling of both local and global dependencies. The consistent improvements across increasing prediction horizons further underscore MAGNN's robustness, scalability, and effectiveness in accurately detecting malicious network traffic.

Table 4 presents the experimental results of the proposed MAGNN framework compared to baseline models, with all results reported to four decimal places. It can be observed that the proposed MAGNN achieves the smallest values for RMSE, MAE, MSE, and MAPE across all four datasets. These lower error values indicate that the predictions made by the MAGNN model are closer to the actual network traffic values, demonstrating its superior accuracy and reliability. The results show that the MAGNN framework is more effective and competitive for malicious network traffic detection tasks compared to other baseline models.

5.3. Analysis of computational efficiency across temporal snapshot sizes

The computational efficiency of the proposed MAGNN model and baseline methods is evaluated by measuring the training duration per epoch across four benchmark datasets. Fig. 9 presents the time required for a single training epoch at varying temporal sizes (300s, 600s, and 900s). As expected, computational cost increases with longer snapshot durations due to greater data volume and complexity.

For the CTU-13 dataset (Fig. 9(a)), MAGNN demonstrates competitive training efficiency. While DE-GNN and TCGNN experience a noticeable increase in training time at 900s, MAGNN maintains a relatively low training duration, highlighting its efficiency. Similarly, for ISCXVPN2016 (Fig. 9(b)), MAGNN strikes an effective balance between performance and training time. Although NEGAT and E-GraphSAGE show slightly lower durations at 300s and 600s, MAGNN scales more efficiently at 900s, outperforming others in both speed and accuracy.

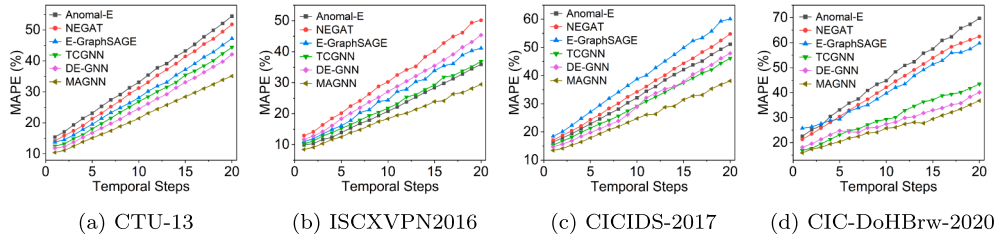
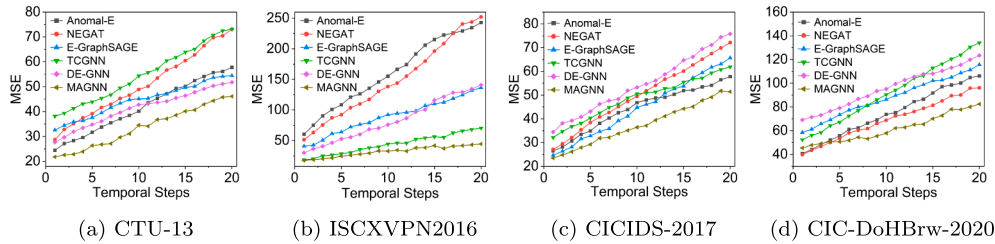
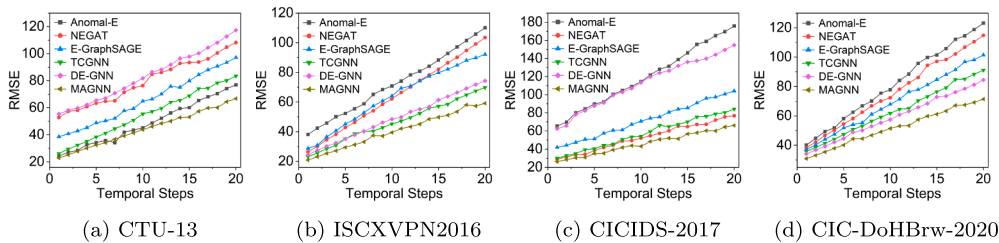
On the CICIDS-2017 dataset (Fig. 9(c)), MAGNN consistently exhibits lower training times than DE-GNN and TCGNN, whose durations increase sharply with larger temporal sizes. This illustrates the architectural efficiency of MAGNN in handling more complex, time-extensive data. For the CIRA-CIC-DoHBrw-2020 dataset (Fig. 9(d)), MAGNN again achieves superior scalability, maintaining the lowest training time at 900s among all compared models.

The observed trends confirm that MAGNN offers both fast convergence and reduced computational cost while sustaining high predictive performance. In contrast, baseline models such as DE-GNN and TCGNN incur substantial overhead at larger temporal sizes. The efficiency of

Table 4

Error-based performance comparison of MAGNN and baseline models across four benchmark datasets.

Data sets	Model	RMSE	MAE	MSE	MAPE
CTU-13	Anomal-E	78.1240	61.2413	6103.2354	11.1023
	NEGAT	75.8967	58.7856	5769.4301	10.4321
	E-GraphSAGE	74.5893	57.6842	5567.8902	10.0516
	TCGNN	73.2384	56.5423	5362.9284	9.7562
	DE-GNN	72.1032	54.9140	5198.1235	9.4213
	MAGNN	69.8452	51.7631	4879.1320	8.9456
ISCXVPN2016	Anomal-E	65.2481	50.9243	4256.9341	9.4212
	NEGAT	63.5674	48.8562	4057.8924	8.7623
	E-GraphSAGE	61.9843	47.2314	3849.4351	8.2345
	TCGNN	60.4821	45.9852	3695.1423	7.9421
	DE-GNN	59.2340	44.8230	3567.1243	7.6512
	MAGNN	56.7123	42.3412	3218.5342	7.1435
CICIDS-2017	Anomal-E	70.7841	55.2384	5012.9843	10.3412
	NEGAT	68.4921	52.8342	4789.1234	9.8745
	E-GraphSAGE	66.9812	50.8921	4489.3214	9.5643
	TCGNN	65.4320	49.2356	4287.2134	9.2342
	DE-GNN	63.8924	47.9841	4056.8942	8.9245
	MAGNN	61.0234	45.2312	3735.1284	8.3456
CIRA-CIC-DoHBrw-2020	Anomal-E	82.3412	66.2314	6194.9834	11.5243
	NEGAT	79.8642	63.8941	5982.3412	10.8723
	E-GraphSAGE	77.2845	61.3212	5723.4501	10.5412
	TCGNN	75.7642	59.8324	5498.3214	10.1123
	DE-GNN	74.1231	58.4210	5287.8912	9.8456
	MAGNN	71.4532	55.2341	4987.4512	9.3412

**Fig. 6.** MAPE of different methods under different prediction temporal steps.**Fig. 7.** MSE of different methods under different prediction temporal steps.**Fig. 8.** RMSE of different methods under different prediction temporal steps.

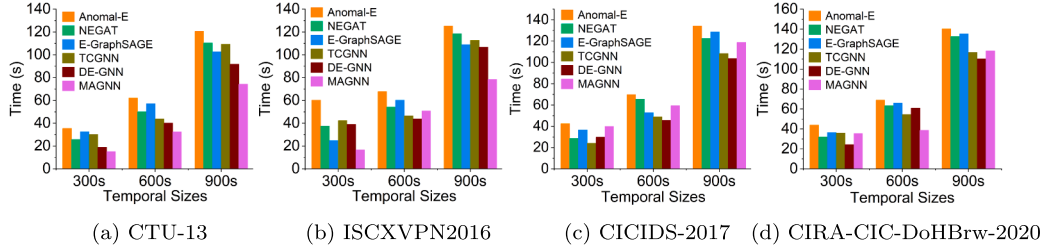


Fig. 9. Duration of a single training epoch when using the different dataset.

MAGNN's design ensures computational viability, making it suitable for deployment in dynamic and large-scale network environments.

5.4. Computation time analysis

In this section, we analyze the computational time overhead of the proposed MAGNN framework compared to the state-of-the-art models. It can be seen from Table 5 that the proposed MAGNN model consistently achieves the lowest training and inference times compared to the baseline models across all datasets. Specifically, the training time for MAGNN is significantly reduced, demonstrating its computational efficiency in handling malicious network traffic detection tasks. For instance, on the CTU-13 dataset, MAGNN achieves a training time of 18.5s per epoch, while the closest competing model, DE-GNN, requires 26.7s per epoch. Similarly, the inference time for MAGNN on the same dataset is 1.4s per epoch, which is notably faster than all baseline models.

The superior computational efficiency of MAGNN can be attributed to the following reasons:

- The multi-scale adaptive architecture of MAGNN efficiently processes both node-level and edge-level information while reducing redundant computations.
- The integrated contrastive learning mechanism is optimized to minimize the computational overhead during both training and inference.
- MAGNN avoids complex recursive operations and computational bottlenecks often encountered in models like TCGNN and DE-GNN.

While models such as E-GraphSAGE and TCGNN perform well in terms of training efficiency, their inference times remain higher compared to MAGNN. For instance, on the CIRA-CIC-DoHBrw-2020 dataset, MAGNN achieves an inference time of 2.4s per epoch, whereas E-GraphSAGE and TCGNN take 4.5s and 4.2s, respectively.

These results highlight the ability of MAGNN to achieve a balance between computational efficiency and detection accuracy. The reduced training and inference times make MAGNN highly suitable for real-time malicious traffic detection, where fast and efficient processing is critical. Additionally, the scalability of MAGNN across multiple datasets further validates its robustness and practical applicability for large-scale cybersecurity systems.

5.5. Comparison of graph construction paradigms

To evaluate the effectiveness of the graph construction strategy, we compare three different paradigms for representing network traffic as graphs, each integrated into the MAGNN framework: (a) a host-centric graph, where nodes represent IPs or devices and edges represent communication flows; (b) a flow-centric graph, where nodes correspond to individual traffic flows and edges capture topological or temporal relationships (e.g., shared hosts or adjacent timestamps); and (c) a feature-level graph, where nodes represent flow features and edges reflect statistical correlations among them.

Table 5

Training and inference time per epoch (in seconds) of MAGNN compared to baseline models across four benchmark datasets.

Dataset	Method	Training (s/epoch)	Inference (s/epoch)
CTU-13	Anomal-E	35.2	3.5
	NEGAT	32.8	3.2
	E-GraphSAGE	30.1	2.9
	TCGNN	28.4	2.6
	DE-GNN	26.7	2.3
	MAGNN	18.5	1.4
ISCVN2016	Anomal-E	40.1	4.1
	NEGAT	37.5	3.8
	E-GraphSAGE	35.2	3.5
	TCGNN	32.8	3.2
	DE-GNN	30.6	3.0
	MAGNN	21.4	1.6
CICIDS-2017	Anomal-E	45.7	4.8
	NEGAT	42.3	4.5
	E-GraphSAGE	39.8	4.2
	TCGNN	37.2	3.8
	DE-GNN	35.4	3.6
	MAGNN	24.5	1.9
CIRA-CIC-DoHBrw-2020	Anomal-E	50.8	5.2
	NEGAT	48.3	4.9
	E-GraphSAGE	44.6	4.5
	TCGNN	42.1	4.2
	DE-GNN	40.4	4.0
	MAGNN	28.7	2.4

Fig. 10 provides a conceptual and performance comparison of the three paradigms across four widely used datasets: CTU-13, ISCVN2016, CICIDS-2017, and CIRA-CIC-DoHBrw-2020. The results clearly show that our host-centric paradigm consistently outperforms the other two across all datasets and evaluation metrics. For instance, on CIRA-CIC-DoHBrw-2020, the host-centric approach achieves an accuracy of 98.91%, compared to 94.61% for the flow-centric model and 90.70% for the feature-centric model. Similar trends are observed across the remaining datasets, with an average improvement of 4-8% in each metric.

These findings validate our design choice to represent devices as nodes and model the communication flow between them. This structure preserves both topological and temporal context, enabling more robust multi-scale behavioral pattern learning. In contrast, while the flow-centric approach captures localized flow relationships, it lacks broader structural awareness. The feature-centric paradigm, although lightweight, struggles to encode sequential dependencies or hierarchical semantics, especially in encrypted or high-volume traffic scenarios.

5.6. Distributed training evaluation

To further validate the scalability and computational efficiency of the proposed MAGNN framework, we conducted a comprehensive evaluation of its training behavior under both non-distributed (single GPU)

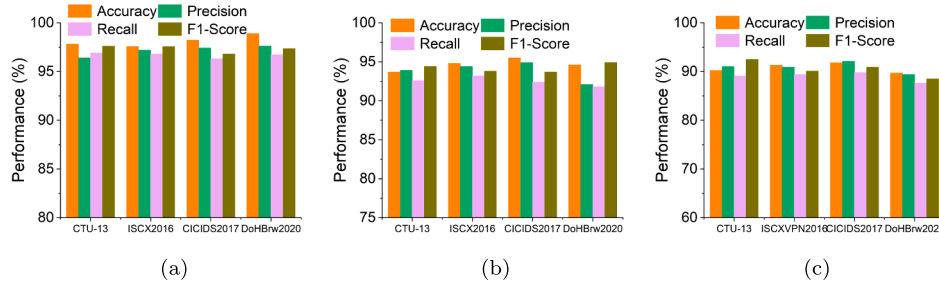


Fig. 10. Comparative performance of different graph construction paradigms on four network traffic datasets: (a) IPs or devices as nodes with communication flows as edges, (b) Traffic flows as nodes with topological relationships as edges, and (c) Flow features as nodes with feature correlation-based edges.

and distributed (multi-GPU) configurations. The distributed implementation employs PyTorch's Distributed Data Parallel (DDP) module across multiple NVIDIA GPUs, enabling parallel gradient computations and synchronized parameter updates. We selected PyTorch DDP due to its communication efficiency and widespread adoption in research and production environments, establishing a reliable baseline for scalable training evaluation.

Figs. 11(a)-11(d) present the normalized distribution of network traffic data (nodes and edges) across 1, 2, 3, and 4 GPUs for the CTU-13, ISCXVPN2016, CICIDS-2017, and CIRA-CIC-DoHBrw-2020 datasets, respectively. The results show that the graph partitioning achieves a stable and efficient distribution of traffic data even as the number of GPUs increases. For instance, on the CTU-13 dataset (Fig. 11(a)), the node and edge traffic distribution remained between approximately 88% and 110% as the number of GPUs increased from 1 to 4, indicating consistent workload sharing. Similar trends are observed for the ISCXVPN2016 (Fig. 11(b)), CICIDS-2017 (Fig. 11(c)), and CIRA-CIC-DoHBrw-2020 (Fig. 11(d)) datasets, demonstrating that MAGNN maintains stable and efficient distribution of network traffic data across GPUs.

Figs. 12(a)-12(d) report the normalized training time and speedup percentages achieved with distributed configurations. For all datasets, the training time at 1 GPU is normalized to 100%, and the relative training times for 2, 3, and 4 GPUs show significant reductions. For instance, on the CTU-13 dataset (Fig. 12(a)), the training time decreases by approximately 39.52% when using two GPUs, and by up to 59.83% with four GPUs. Similar improvements are observed on the ISCXVPN2016 (Fig. 12(b)), CICIDS-2017 (Fig. 12(c)), and CIRA-CIC-DoHBrw-2020 (Fig. 12(d)) datasets, with speedup rates consistently exceeding 50% when utilizing three or more GPUs.

Importantly, distributed training did not compromise detection performance. The convergence rates and classification metrics remained comparable between single-GPU and multi-GPU settings, ensuring that the parallelization process preserved model effectiveness. The workload variations observed during graph partitioning did not adversely impact model convergence or stability.

These results demonstrate that MAGNN not only achieves high detection accuracy but also scales efficiently with additional computational resources. This capability makes MAGNN a practical and robust solution for large-scale, real-time network traffic detection scenarios, where rapid model updates and the ability to handle massive graph structures are essential.

5.7. Ablation study

The effectiveness of the proposed MAGNN framework was analyzed by conducting an ablation study, systematically disabling specific modules to assess their contributions to overall performance. Table 6 presents the results, demonstrating the critical role of each component in achieving optimal results for malicious network traffic detection.

The complete MAGNN Model consistently outperformed its ablated variants, achieving the highest accuracy (98.45%) and F1-score

Table 6

Ablation study of the proposed MAGNN model.

Ablation Component	Acc.	F1-Score	MSE	RMSE	MAE	MAPE
Full Model (MAGNN)	98.45	97.58	0.06	0.20	0.31	1.08
Without Multi-Head Contrast	97.30	96.85	0.09	0.25	0.42	1.15
Without Edge-Level Contrast	97.10	96.45	0.10	0.26	0.48	1.22
Without Temporal-Node Contrast	96.80	96.00	0.12	0.28	0.50	1.25
Without Graph Augmentation	96.50	95.75	0.13	0.30	0.54	1.28
Without Integrated Loss	96.90	96.20	0.11	0.27	0.49	1.20

(97.58%) across all metrics. Additionally, the full model recorded the lowest error values for MSE, RMSE, MAE, and MAPE, highlighting its superior predictive capability and robustness.

When the multi-head hierarchical contrast module was removed, the performance dropped significantly, with a reduction in accuracy (97.30%) and F1-score (96.85%), indicating that this component is crucial for capturing multi-scale hierarchical dependencies within the graph. Similarly, disabling the edge-level contrast resulted in further degradation of performance, particularly in error metrics, with MSE and RMSE increasing to 0.10 and 0.26, respectively. This underscores the importance of edge embeddings in capturing packet-level relationships for malicious traffic detection.

The temporal-node contrast, designed to model temporal dependencies in network traffic, also proved essential. Its removal led to further performance reductions, with an F1-score dropping to 96.00%, emphasizing the role of temporal information in identifying evolving threats and malicious behaviors. The absence of the graph augmentation module, which incorporates edge modifications and random walk sampling, resulted in the most significant degradation in error metrics, with MAE and MAPE increasing to 0.54 and 1.28, respectively. This highlights the importance of augmentation strategies in improving model generalization and robustness to diverse network scenarios.

Finally, replacing the integrated loss function with individual loss objectives revealed the importance of the multi-scale optimization framework. The performance of the model decreased across all metrics, with accuracy dropping to 96.90% and F1-score to 96.20%. This demonstrates that the integrated loss function enables the framework to effectively combine hierarchical, temporal, and edge-level information for optimal performance.

Finally, the ablation study validates the design choices in the MAGNN framework, with each component contributing significantly to the overall performance. The results highlight the effectiveness of combining multi-scale adaptive graph learning, contrastive mechanisms, and augmentation strategies for malicious network traffic detection.

5.8. Adaptability to evolving threats

MAGNN is inherently designed to enhance generalization through contrastive learning with graph augmentation and hierarchical representations, which can help improve resilience to zero-day attacks and

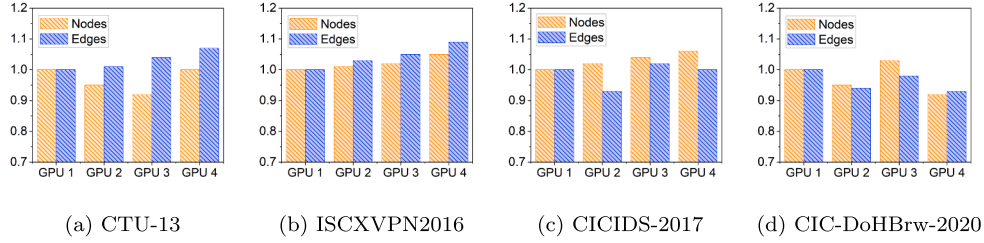


Fig. 11. MAGNN achieves stable and efficient workload distribution across multiple GPUs, with node and edge partitioning.

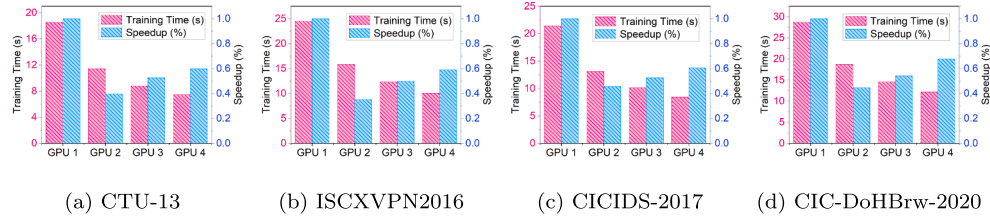


Fig. 12. MAGNN achieves substantial training efficiency gains as the number of GPUs increases, demonstrating its scalability under distributed training setups.

unforeseen anomalies. Specifically, the multi-scale contrastive framework enables the model to learn robust structural and semantic patterns rather than relying solely on known attack signatures. This capability supports the detection of zero-day threats that exhibit novel behaviors.

While MAGNN does not explicitly model long-term temporal dependencies associated with Advanced Persistent Threats (APTs), its hierarchical and edge-level encoding can capture fine-grained interactions that may indicate dormant or stealthy patterns. Additionally, the use of stochastic graph augmentations (e.g., edge modifications and subgraph sampling) promotes adaptability to dynamic changes in traffic feature distributions.

Nonetheless, we acknowledge that further improvements could be achieved by integrating online learning mechanisms or memory-based architectures to track evolving behaviors over time. Future work may explore continual contrastive learning or temporal memory networks to enhance MAGNN's responsiveness to concept drift and long-term attack evolution.

6. Conclusion

The proposed MAGNN framework presents significant advancements in malicious network traffic detection by leveraging multi-scale adaptive graph learning and contrastive mechanisms. By integrating temporal-node, edge-level, and hierarchical contrastive learning, MAGNN effectively captures both local and global dependencies within network traffic data, ensuring robustness and scalability across diverse environments. Experimental evaluations on four benchmark datasets-CTU-13, ISCXVPN2016, CICIDS-2017, and CIRA-CIC-DoHBrw-2020-demonstrate the consistent superiority of MAGNN over state-of-the-art baseline models, including Anomal-E, NEGAT, E-Graph SAGE, TCGNN, and DE-GNN. MAGNN achieved higher accuracy and F1-scores while maintaining lower error rates across all datasets, confirming its ability to handle complex and evolving traffic patterns. The use of a novel graph augmentation strategy and a self-supervised learning paradigm further reduced dependency on labeled data, enhancing its applicability to real-world cybersecurity challenges.

The results highlight MAGNN's robustness, faster convergence, and computational efficiency compared to existing approaches. Its strong generalization capability across heterogeneous network environments and its effectiveness in detecting emerging threats position it as a practical and scalable solution for large-scale cybersecurity systems. Looking ahead, several promising directions exist for extending this

work. First, incorporating advanced graph augmentation techniques and dynamic graph representations could further enhance adaptability to rapidly changing network conditions. Integrating additional contextual features-such as application-layer protocols or user behavior metrics-may improve the detection of previously unseen attack patterns.

Expanding evaluations to include larger, more diverse datasets and real-time network traffic would further validate the framework's practical applicability. Exploring hybrid models that combine graph-based learning with other paradigms, such as transformers or reinforcement learning, also holds potential. Lastly, addressing explainability in graph neural networks remains critical; developing interpretable mechanisms within MAGNN will be key to fostering trust and adoption in operational cybersecurity environments. These future directions aim to further refine MAGNN's capabilities, ensuring its continued effectiveness in the face of evolving network threats.